

Librika Platform User Manual

Complete Technical Specifications, Functionalities & API Call Flows

System Version:	v3.1.2 (LMS/ILMS Production Release)
Database Engine:	SQLite 3 (WAL Mode Enabled)
AI Completions:	NVIDIA Nemotron completions API (30B parameter reasoning)
Security Level:	Role-Based Access Control (RBAC)
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Chapter 1: Platform Overview & System Architecture

Librika (e-Pathshala) is a next-generation integrated school library catalog system (ILMS) and reading engagement platform. The system operates on a lightweight B2B multi-tenant setup where database isolation and subscription allowances enforce strict data separation between school instances.

A. Tech Stack Specifications

- **Application Framework:** Flask (Python 3.9+) with session-based route security filters.
- **Local Storage:** SQLite 3 database engines. To handle high concurrency, Write-Ahead Logging (WAL) is enabled by executing `PRAGMA journal_mode=WAL`, allowing parallel reads during active writes.
- **AI Cognitive Layer:** NVIDIA integrate completions gateway. Leveraging the Nemotron-3-nano model for structuring OCR context and generating reading assessments.
- **Design system:** Responsive HTML5 markup with vanilla CSS, incorporating 3D transformations for payment interfaces.

B. Database Schema Definitions

The system maintains two database structures (`demo.db` and `library_v3.db`) with identical schemas:

1. **schools:** Tracks license subscription (FREE, BASIC, PROFESSIONAL), expiry dates, and registration limits.
2. **books:** Physical assets containing title, author, `barcode_id`, `isbn`, publisher, quantity, and categories.
3. **digital_content:** Digital e-Library uploads (PDF/EPUB path logs, cover URLs, contributors).
4. **transactions:** Circulation logs recording checkouts, due dates, returned dates, and point penalties.
5. **quiz_attempts:** Passing metrics, scores, timestamps per student per quiz.

Chapter 2: Core Modules & Detailed Call Flows

Module A: AI Smart Book Scanner

Integrates a parallel client-side scanning interface that processes book cover photos, extracts metadata via LLM vision models, and auto-saves books. If a scanned book already exists, the system automatically increments the catalog copy stock instead of throwing errors.

Step	Call Component / Endpoint	Action & Output Details
1	POST /admin/api/upload-cover	Takes captured camera frame. Resizes image under 600px width (for DB base64 compliance) and saves file. Returns static path URL.
2	POST /api/scan-vision	Passes the cover path. Encodes to base64, queries NVIDIA Nemotron Vision API to extract title, publisher, class, and ISBN. Strips LLM reasoning thought blocks before JSON load.
3	API Fallback 1: Google Books	If VLM returns empty critical fields, requests Google Volumes API for catalog data using title query (e.g. isbn:123).
4	API Fallback 2: DDG Web Search	If Google fails (NCERT/local books), queries DuckDuckGo for snippets. Passes crawled results to LLM for metadata correction.
5	POST /admin/api/save-scanned	Checks database. If unique ISBN/Title exists, calls /admin/api/add-copy to increment copies. Else, inserts new record and generates Code-128 barcode image.

Module B: Reader Leaderboard System

Fosters reading motivation by computing scores based on user responsibility and comprehension quiz passes. Splits rankings into Physical, Digital, and Overall Champion lists.

Step	Call Component / Endpoint	Action & Output Details
1	Issue Book (Physical/Digital)	Allocates +5 points to student score and inserts transactions record.
2	Return Book (Librarian return)	Evaluates returned date. If before due date: awards +15 points. If late: deducts -20 points. If lost: deducts -50 points.
3	Submit Review	Writes user review record. Once approved by admin moderation: awards +20 points.
4	AI Quiz Completion	Checks answers. If passed: awards +50 points to leaderboard totals.

Module C: AI Quiz Comprehension System

To verify students read and comprehend borrowed materials, the platform dynamically generates quizzes using NVIDIA Nemotron. Strict anti-exploit blocks are enforced to prevent points farming.

Step	Call Component / Endpoint	Action & Output Details
1	GET /student/quiz//	Evaluates quiz eligibility parameters. Locks quiz if minimum reading period is not met based on page length (Short: 2d, Med: 5d, Long: 7d). Checks 90-day cooldown to block duplicate points farming.
2	ai_generate_quiz()	If quiz record does not exist in book_quizzes table, sends book metadata to NVIDIA completions API. Formats a 5-question multiple-choice JSON array.
3	POST /student/quiz//	Submits student answers. Grades attempt out of 5. Logs pass/fail metrics. If passed, awards +50 points to student profile and locks future attempts.

Module D: AI Semantic Search Engine

Replaces string matching with semantic context understanding. Toggling  **AI Search** filters books by concept relevance rather than literal matches.

Step	Call Component / Endpoint	Action & Output Details
1	GET /student/browse?q=[query]&ai=true	User checks AI toggle and submits search. Intercepts request in Flask. Queries database for available catalogs.
2	perform_ai_semantic_search()	Constructs prompt containing query and book details. Passes parameters to completions endpoint to calculate relevance score (0-100) for each item ID.
3	Render student_browse.html	Filters items scoring 0. Sorts list by match percentage descending and renders a gradient  XX% Match badge on card overlays.

Module E: Subscriptions & Billing Portal

Manages plan upgrades, limit verification, and payment simulations.

Step	Call Component / Endpoint	Action & Output Details
1	Upgrade Select	Admin opens /billing. System reads limits and lists starter, basic, and professional packages.
2	Checkout Form Click	Pops up the 3D payment modal. Intercepts inputs to update card template. Flips 180 degrees on focusing CVV input.

3	POST /billing/checkout	Simulates transaction authorization, creates invoice logs, sets subscriptionStatus to 'active', and upgrades catalog quotas in SQLite.
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